

Weather-Aware Itinerary Planning: From Solver-Backed Routes to Inspectable Decision Support

Literature review progress, validation boundaries, and TRB vs. IUI/CHI positioning

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Meeting with Prof. Choi

Core framing: **feasible itinerary generation** + **human-understandable tradeoff revision**

Talk roadmap

What I reviewed

- ▶ Tourist Trip Design / Orienteering
- ▶ Personalized and context-aware itinerary recommendation
- ▶ LLM travel planning, symbolic validation, and LLM-as-parser systems
- ▶ Disruption-aware replanning
- ▶ HCI, explainability, mixed initiative, and plan evaluation

What I need feedback on

- ▶ First paper framing: TRB vs. IUI/CHI
- ▶ Which RQ should be central
- ▶ Whether language/agent functions should stay future work
- ▶ What empirical evidence is realistic

The goal of this presentation is to narrow a broad prototype into a publishable research story.

Main takeaway: the contribution should shift

Weak framing

“Weather + Gurobi itinerary optimizer”

- ▶ Too close to existing TTDP / routing work
- ▶ Hard to claim algorithmic novelty
- ▶ Does not fully use the dashboard contribution

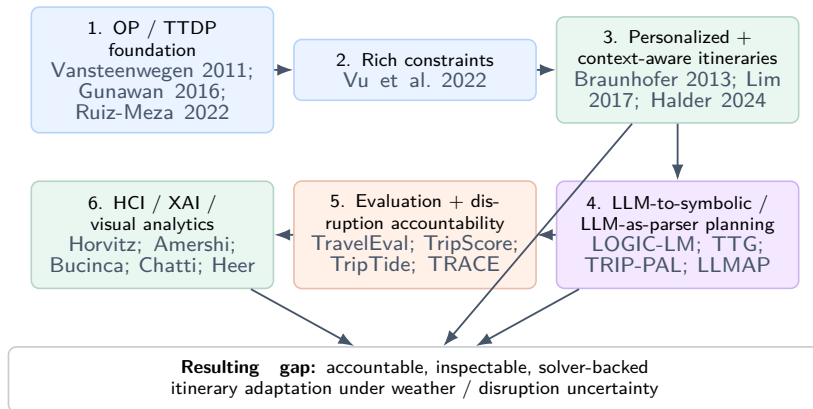
Stronger framing

Inspectable, solver-backed decision support

- ▶ Feasible route alternatives
- ▶ Weather / disruption adaptation
- ▶ Explanations and user revision
- ▶ Provenance, uncertainty, and solver status

The publishable gap is not only computing a plan, but helping users **understand, compare, and revise** the plan.

Progressive literature ladder: how the project angle emerged



Recent arXiv/demo papers are useful for framing, but final publication status and scope limits must be verified before citation.

Literature block 1: TTDP says this is not a ranked-list task

Core problem

- ▶ Select a subset of POIs
- ▶ Sequence them into feasible days
- ▶ Respect time, distance, budget, and lodging constraints
- ▶ Maximize collected trip value

Vansteenwegen et al. 2011; Gunawan et al. 2016;
Ruiz-Meza and Montoya-Torres 2022

Implication for my project

- ▶ Do not claim novelty for basic POI selection + routing
- ▶ Use Tourist Trip Design Problem (TTDP) as the formal foundation
- ▶ Focus novelty on weather adaptation, route alternatives, and user-facing inspection

A high-rating attraction is not necessarily good once travel time, daily budget, weather, and hotels are considered.

Literature block 2: rich constraints create hidden feasibility failures

Vu et al. 2022: Branch-and-check

- ▶ Master problem selects candidate locations
- ▶ Subproblem checks schedule feasibility
- ▶ Infeasible selections generate cuts
- ▶ Rich constraints include mandatory visits, categories, order rules, budgets, time windows

Project design lesson

- ▶ Feasibility failures should not stay hidden inside the solver
- ▶ The dashboard should explain **why a candidate fails**
- ▶ “Why skipped” can be generated by attempted insertion or repair checks



Literature block 3: personalization still needs feasibility

Personalized itinerary recommendation

- ▶ User interest and POI utility are necessary
- ▶ Context matters: time, queues, weather, crowding, seasonality
- ▶ But personalization alone can create infeasible routes

Braunhofer et al. 2013; Lim et al. 2017; Halder et al. 2024

How it maps to my system

- ▶ Interest profiles and social / must-go signals estimate value
- ▶ Weather risk and detour cost modify realized suitability
- ▶ The solver decides which high-value POIs(Places of Interest) can actually fit

The route should expose the conflict between **what the traveler likes** and **what the trip can support**.

Literature block 4: LLM planning shows why validation boundaries matter

Problem in direct LLM travel planning

- ▶ Plans can be fluent but infeasible
- ▶ Common failures: spatial, temporal, budget, lodging, opening-hour, and constraint violations
- ▶ Surface plausibility is not enough

TravelPlanner; TravelEval; TripScore

Parser + solver direction

- ▶ Use LLMs to interpret preferences, constraints, and disruptions
- ▶ Compile confirmed meaning into structured planning inputs
- ▶ Let planner / solver verify encoded feasibility
- ▶ Show translation, solver/fallback, and evaluation status to users

LOGIC-LM; To the Globe / TTG; TRIP-PAL; LLMAP

This supports a language/agent future direction, but the LLM should **formulate**; it should not silently become the final route authority.

Literature block 5: TripTide gives the disruption-repair vocabulary

Disruption types

- ▶ transport
- ▶ accommodation
- ▶ restaurant
- ▶ attraction
- ▶ weather / fatigue / other

Severity levels

- ▶ step-level: one POI
- ▶ day-level: one day
- ▶ plan-level: whole itinerary

Traveler tolerance

- ▶ Flexi-Venturer: accepts rerouting
- ▶ Plan-Bound: wants minimal changes
- ▶ Evaluation: preservation of intent, responsiveness, adaptability

My project can turn weather risk into a disruption signal, then use optimization to verify repairs rather than relying on LLM-only rewriting.

TripTide

2025, treated as recent disruption-aware travel planning work

Literature block 6: HCI/XAI turns routes into inspectable decisions

Design principles from HCI

- ▶ Mixed initiative: system proposes, human controls
- ▶ Make system capabilities and limits visible
- ▶ Support correction, comparison, and feedback
- ▶ Avoid blind overreliance on AI outputs

Horvitz 1999; Amershi et al. 2019; Bucinca et al. 2021

Explanation design

- ▶ Why selected?
- ▶ Why skipped?
- ▶ What changed?
- ▶ What would need to change?
- ▶ What evidence / uncertainty supports this route?

Tintarev and Masthoff; Chatti et al. 2024;
CityHood 2025; TRACE 2026

Current system: what is already useful

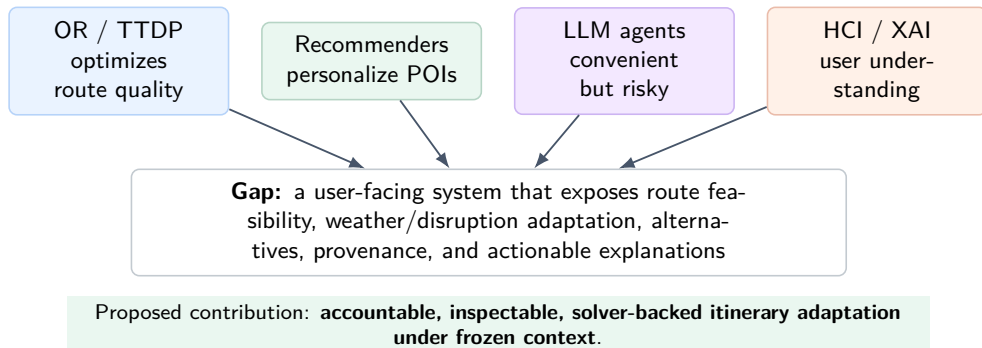
Implemented or partly implemented

- ▶ multi-day California route artifacts
- ▶ POI scoring and interest profiles
- ▶ hotels / base-city decisions
- ▶ route alternatives and trip-length alternatives
- ▶ weather-risk fields and map layers
- ▶ Gurobi-backed route components where available
- ▶ heuristic repair / fallback outputs
- ▶ source confidence and validation artifacts
- ▶ basic why-selected explanation text

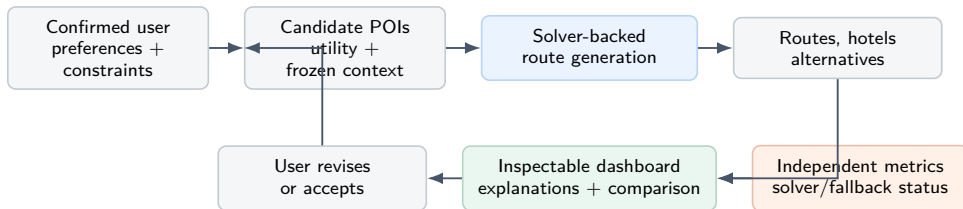
Claim boundary

- ▶ Do not say every route is globally optimal
- ▶ Separate solver-backed from heuristic routes
- ▶ Browser-side interest controls may be preview-only
- ▶ Proxy values need uncertainty labels
- ▶ No user study yet proves trust or decision quality
- ▶ No canonical independent complete-plan evaluator yet

Feasible output alone does not make a route understandable or revisable.



Proposed system framing



Contribution statement

A route-optimization system that turns confirmed inputs, solver output, and evaluation evidence into **inspectable evidence** for traveler decision-making.

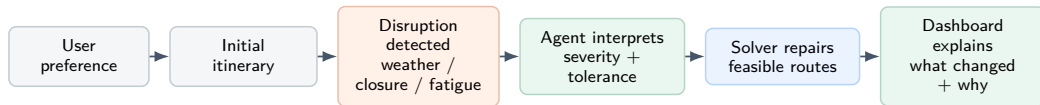
Possible research questions

Primary RQ

How can an **inspectable, solver-backed travel-planning interface** help users understand, compare, and revise multi-day itinerary tradeoffs under weather and disruption uncertainty?

1. **Interpretation:** When preferences or disruptions are translated into planning inputs, what should require user confirmation?
2. **Explanation:** Do why-selected, why-skipped, and what-would-need-to-change explanations improve constraint understanding?
3. **Revision:** Does original-vs-adapted comparison help users make feasible revisions while preserving trip intent?
4. **Reliance:** Do provenance, uncertainty, and solver/fallback labels improve calibrated reliance rather than simply increasing trust?

Agent extension: useful, but should be bounded



Agent should do

- ▶ parse requests into auditable fields
- ▶ classify disruption type and severity
- ▶ infer repair scope: stop / day / whole plan
- ▶ translate traveler tolerance into constraints
- ▶ compare feasible repair alternatives

Agent should not do

- ▶ directly invent final itineraries
- ▶ bypass feasibility checking
- ▶ emit arbitrary solver code or hidden weights
- ▶ silently change hotels or must-go stops
- ▶ hide uncertainty or fallback status

How to operationalize “preservation of intent”

Optimization idea

$$\max U(\text{route}) - \lambda_W R_W - \lambda_T T - \lambda_D \Delta(\text{original})$$

- ▶ utility and must-go value
- ▶ weather exposure penalty
- ▶ travel burden penalty
- ▶ deviation from original itinerary

Deviation can include

- ▶ removed or replaced POIs
- ▶ moved stops between days
- ▶ sequence changes
- ▶ hotel/base-city changes
- ▶ extra driving or cost
- ▶ lost semantic balance: nature, city, culture, food

This turns “preserve traveler intent” from a vague goal into a measurable objective and evaluation metric.

Evaluation plan: computational + human-centered

Independent route metrics

- ▶ feasibility violations
- ▶ utility and selected POIs
- ▶ travel time, cost, and lodging changes
- ▶ weather exposure
- ▶ must-go retention
- ▶ preservation-of-intent score
- ▶ runtime, solver status, fallback status

TravelEval and TripScore-style complete-plan evaluation

User-study metrics

- ▶ constraint-comprehension accuracy
- ▶ ability to explain why a stop was skipped
- ▶ revision quality and feasibility
- ▶ perceived control
- ▶ workload and satisfaction
- ▶ calibrated reliance, not just higher trust

HCI / XAI evaluation: Amershi, Bucinca, Knijnenburg, Chatti

Venue positioning: TRB vs. IUI or CHI

	TRB direction	IUI or CHI direction
Core angle	Transportation / operations research decision support	Intelligent interface / human-AI decision support
Emphasis	Route feasibility, weather disruption, robustness, travel time, lodging	Explanations, mixed initiative, user control, comparison, calibrated reliance
Best evidence	Computational baselines, sensitivity analysis, practical mobility implications	User study, task design, interaction logging, comprehension and revision measures
Best RQ	How does weather-aware solver-backed adaptation improve robust tourist mobility?	How does an inspectable interface help users understand and revise solver-backed itineraries?
Risk	May look less transportation-central if too interface-heavy	Needs empirical HCI evidence, not only a dashboard demo

Current intuition: **IUI or CHI** is more distinctive if a focused user study is feasible;
TRB is safer if the first paper emphasizes computational route adaptation.

Near-term prototype for the first study

Controlled scenario

- ▶ one 7-day California itinerary
- ▶ one explicit weather disruption
- ▶ original route + 2–3 adapted alternatives
- ▶ consistent evaluator for all routes

Explanation panels

- ▶ why selected
- ▶ why skipped
- ▶ what changed
- ▶ what would need to change

Implementation cleanup

- ▶ freeze scenario inputs and context snapshots
- ▶ separate planner run, plan artifact, and evaluation report
- ▶ compare all methods using the same metrics
- ▶ label solver-backed vs. heuristic outputs
- ▶ expose proxy assumptions and confidence
- ▶ avoid post-solve route mutation without new evaluation
- ▶ add interaction logging for study mode

Possible discussion questions

1. Is the first contribution stronger as **context-aware route repair** or as an **inspectable decision-support interface**?
2. Which context types should be in the first paper: weather only, or weather plus closures, travel-time disruption, and pace/accessibility?
3. Is user-confirmed structured input acceptable for the first study, leaving LLM parsing as future work?
4. What baseline would be most convincing: context-blind solver, heuristic repair, full replanning, or LLM-only planning?
5. What evidence would make the paper credible for TRB versus IUI/CHI?

Proposed near-term focus: **frozen scenario + solver-backed repair + independent evaluation + contrastive explanations**